

Step 12: Technical Architecture

BookNest — Architectural Diagram (Technology Stack View)

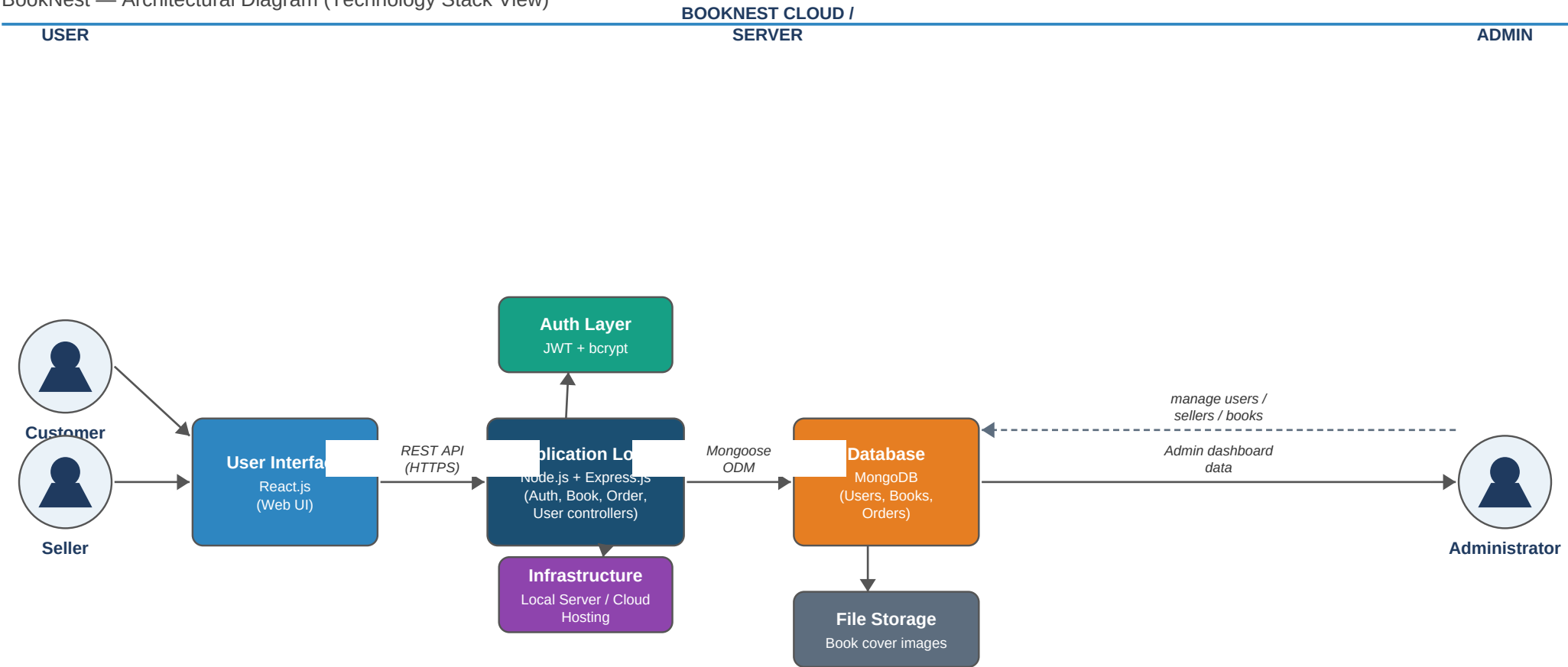


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application (Web UI for Customer, Seller, Administrator)	React.js, HTML, CSS, Bootstrap
2.	Application Logic-1	Authentication and authorization logic for all roles	Node.js, Express.js, JWT
3.	Application Logic-2	Business logic for book browsing, cart, and order processing	Node.js, Express.js (REST controllers)
4.	Application Logic-3	Business logic for seller listing management and admin oversight	Node.js, Express.js (REST controllers)
5.	Database	Data storage for users, books, orders, and sellers	MongoDB (NoSQL), Mongoose ODM
6.	Cloud Database	Managed database hosting service	MongoDB Atlas
7.	File Storage	Storage for book cover images and related assets	Local filesystem / cloud image storage service
8.	External API-1	Payment processing for order checkout (if integrated)	Payment gateway API (e.g. Razorpay/Stripe) — optional/future scope
9.	External API-2	Not used in the current scope of this individual project	Not applicable
10.	Machine Learning Model	Not applicable — BookNest does not use a machine learning component	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on local system or cloud Local Server Configuration: Node.js runtime on developer machine Cloud Server Configuration: Cloud hosting service for frontend and backend	Local, and/or cloud hosting (e.g. Render, Vercel, Netlify)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks and libraries used to build the application	React.js, Node.js, Express.js, Mongoose
2.	Security Implementations	Security and access controls implemented across the application	JWT authentication, bcrypt password hashing, role-based access control, CORS policy
3.	Scalable Architecture	Justification for the scalability of the architecture (3-tier, microservices, etc.)	3-tier architecture: Presentation (React.js) — Application (Node.js/Express.js) — Data (MongoDB), with modular REST API routes
4.	Availability	Design consideration for the availability of the application (e.g. load balancers, distributed servers)	Managed hosting for frontend/backend with database uptime handled by MongoDB Atlas; no dedicated load balancer implemented at this project stage
5.	Performance	Design consideration for the performance of the application (requests per second, caching, CDN)	Indexed MongoDB queries, pagination for large book/order lists, minimal JSON payloads; no CDN implemented at this project stage